DO NOW — RECALL

Dijkstra: shortest paths I

Name: _____ Class: _____ Date: _____

1. Why does quicksort exclude the pivot from its recursive calls?

2. Define a weighted edge.

3. What is the first index of an array?

4. A–B costs 3 and B–C costs 5. What is cost of A→B→C?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

The pivot is already in its final sorted position.

2. Model answer

A connection between vertices with an associated cost or distance.

3. Model answer

0.

4. Model answer

8.

Vocabulary — know these terms

tentative distance — Best known cost to an unsettled vertex.

settled — Vertex whose shortest distance has been finalised.

relaxation — Improve a neighbour's distance via the current vertex.

predecessor — Previous vertex on the best-known route.

non-negative weight — Edge cost of zero or greater.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Reveal and explain

From O, N improves from 5 to 3 and P gets a tentative 10.

From N, P improves to 6; P is next. Route M→O→N→P costs 6.

IMPROVE: Explain one correction to your own reasoning.

NEW LEARNING

Read, annotate and explain

Initialise from the start

Distance(A)=0, all others infinity. Relax A's edges to set B=4 and C=2.

Always choose smallest tentative distance

C=2 settles before B=4. Through C, B improves to 3.

Keep updating neighbours

Settling B changes D from 10 to 8; settling D changes E from 12 to 10.

Reconstruct from predecessors

E←D←B←C←A. Reverse these arrows for A→C→B→D→E, total distance 10.

Dijkstra's requirement

Edge weights must be non-negative. A visited/settled node's shortest distance is then final.

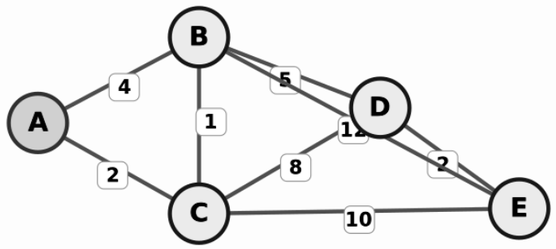
Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

START AT A



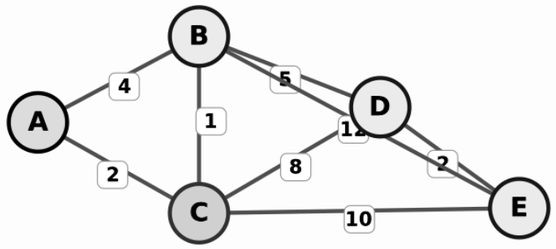
Node	Dist	Previous
A	0	—
B	4	A
C	2	A
D	∞	—
E	∞	—

Distances after settling A

START AT A: graph weighted edges and tentative distances; Distances after settling A

New Learning

SETTLE C (2)



Node	Dist	Previous
A	0	—
B	3	C
C	2	A
D	10	C
E	12	C

B improved from 4 to 3

SETTLE C (2): graph weighted edges and tentative distances; B improved from 4 to 3

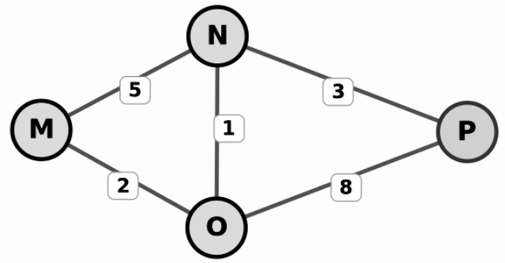
We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find the shortest M→P route using non-negative weights.

From M, set N=5 and O=2; O is the next vertex settled.

REACH P



Node	Dist	Previous
M	0	—
N	3	O
O	2	M
P	6	N

M→O→N→P = 6

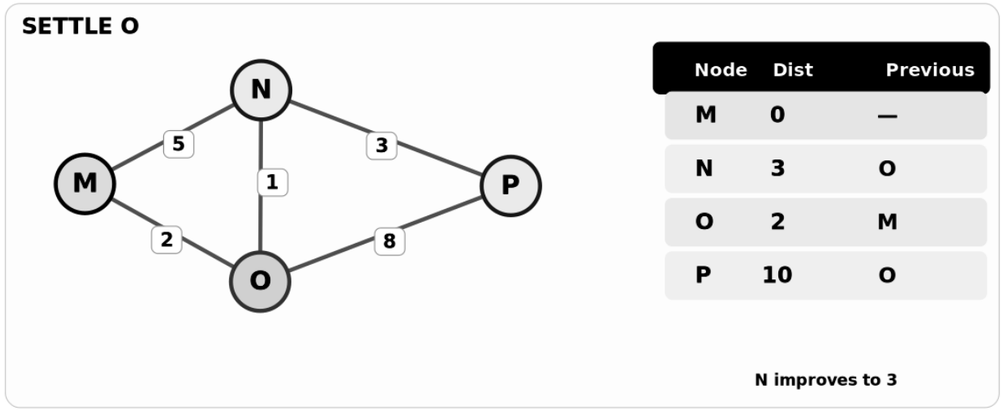
REACH P: graph weighted edges and tentative distances; M→O→N→P = 6

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find the shortest M→P route using non-negative weights.

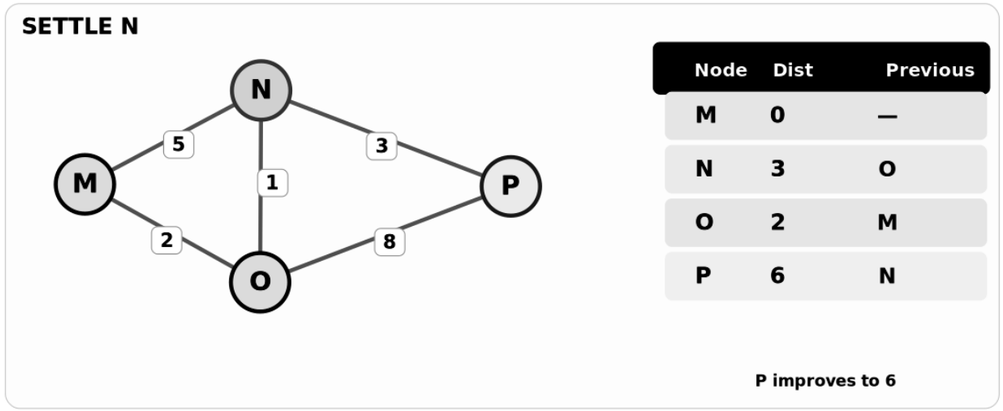
From M, set N=5 and O=2; O is the next vertex settled.



SETTLE O: graph weighted edges and tentative distances; N improves to 3

Find the shortest M→P route using non-negative weights.

From M, set N=5 and O=2; O is the next vertex settled.

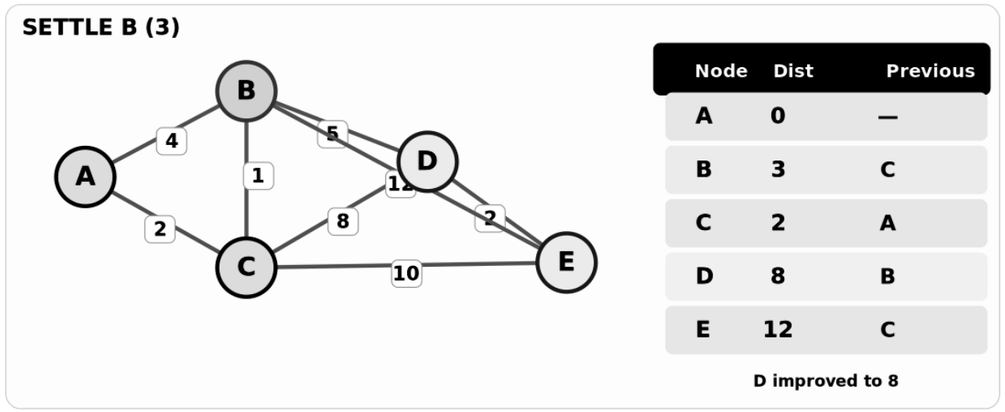


SETTLE N: graph weighted edges and tentative distances; P improves to 6

New Learning · original lesson diagrams

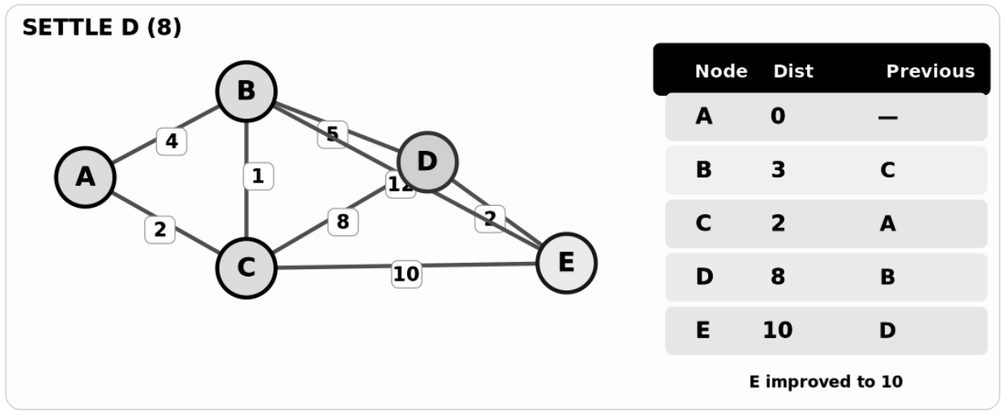
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



SETTLE B (3): graph weighted edges and tentative distances; D improved to 8

New Learning



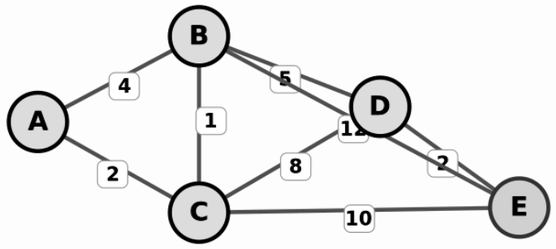
SETTLE D (8): graph weighted edges and tentative distances; E improved to 10

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

REACH E (10)



Node	Dist	Previous
A	0	—
B	3	C
C	2	A
D	8	B
E	10	D

Route A→C→B→D→E = 10

REACH E (10): graph weighted edges and tentative distances; Route A→C→B→D→E = 10

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Dijkstra always chooses the next vertex by its smallest edge weight.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why Dijkstra must not be relied upon when negative-weight edges are possible, and contrast tentative and settled distances.

I DO — WORKED EXAMPLE

Follow the worked method

Find the shortest P→S route using non-negative weights.

Predict the next state

P gives tentative Q=3 and R=7.

Reveal and explain

Settle Q=3, improving R to 5 and discovering S=9.

Reveal and explain

Settle R=5, improving S to 6.

Reveal and explain

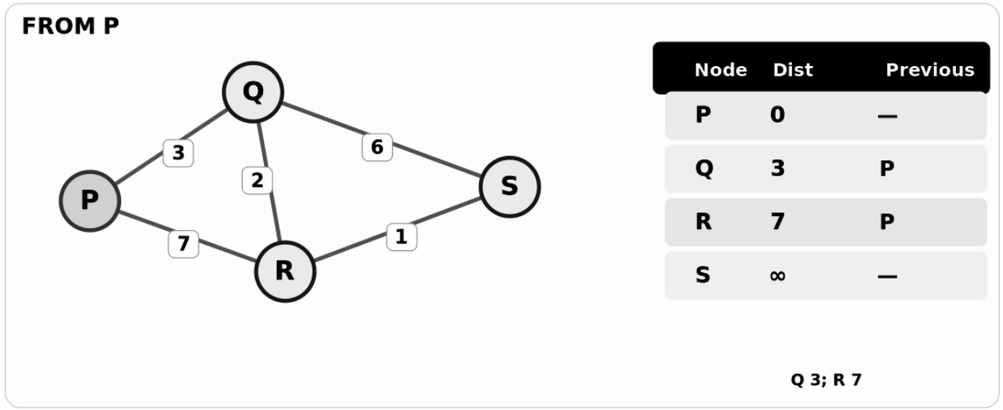
Predecessors S←R←Q←P give P→Q→R→S, cost 6.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find the shortest P→S route using non-negative weights.

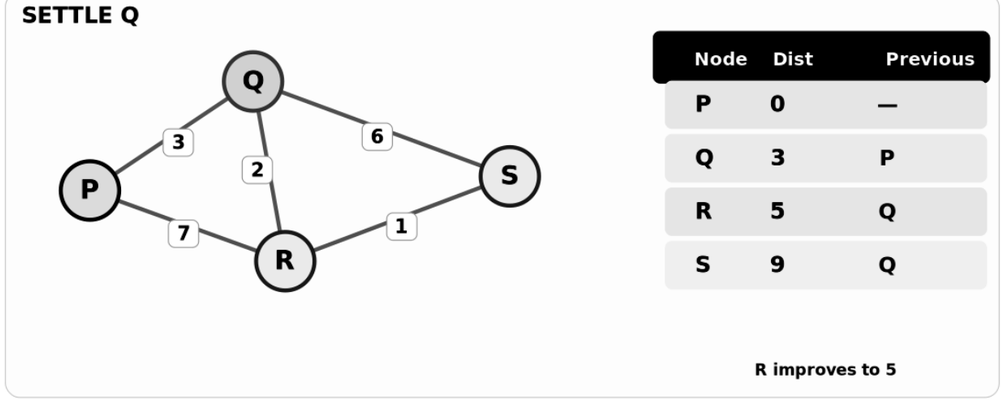
P gives tentative Q=3 and R=7.



FROM P: graph weighted edges and tentative distances; Q 3; R 7

Find the shortest P→S route using non-negative weights.

P gives tentative Q=3 and R=7.



SETTLE Q: graph weighted edges and tentative distances; R improves to 5

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

A–B=4 and A–C=2. Starting Dijkstra at A, state the tentative distances of A, B and C after relaxing A's neighbours.

Question 2

On edges S–A=3, S–B=7, A–B=2, B–T=2 and A–T=8, find the shortest S→T path with full distance reasoning.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages.
DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict the next state From M, set N=5 and O=2; O is the next vertex settled.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

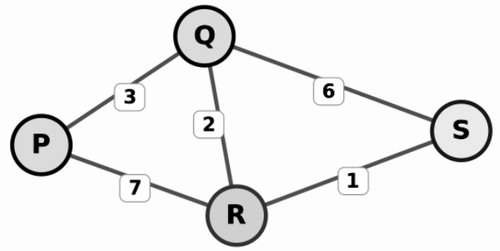
I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find the shortest P→S route using non-negative weights.

P gives tentative Q=3 and R=7.

SETTLE R



Node	Dist	Previous
P	0	—
Q	3	P
R	5	Q
S	6	R

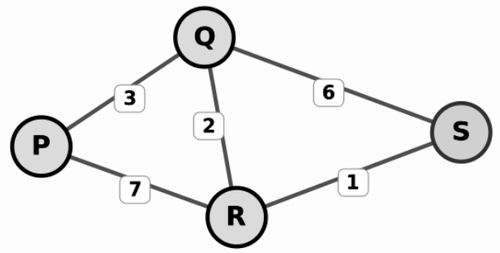
S improves to 6

SETTLE R: graph weighted edges and tentative distances; S improves to 6

Find the shortest P→S route using non-negative weights.

P gives tentative Q=3 and R=7.

REACH S



Node	Dist	Previous
P	0	—
Q	3	P
R	5	Q
S	6	R

P→Q→R→S = 6

REACH S: graph weighted edges and tentative distances; P→Q→R→S = 6

WE DO — GUIDED PRACTICE

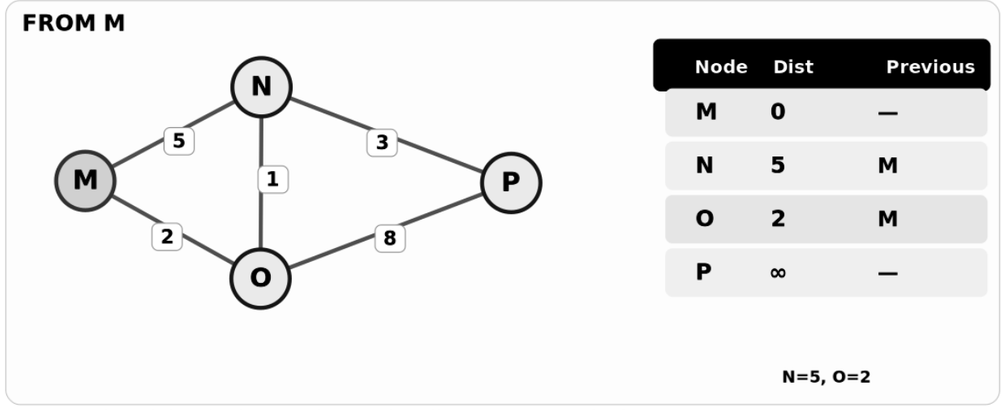
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find the shortest M→P route using non-negative weights.

From M, set N=5 and O=2; O is the next vertex settled.



Node	Dist	Previous
M	0	—
N	5	M
O	2	M
P	∞	—

FROM M: graph weighted edges and tentative distances; N=5, O=2